

The Apple Does Not Fall Far From the Tree: Intergenerational Persistence of Dietary Habits *

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Abstract

This paper provides novel evidence on how dietary habits – a key health behavior – are transmitted across generations, exploiting unique grocery transaction data linked with administrative records. We document strong intergenerational persistence in dietary habits, exceeding that of income, and consider several channels that might explain this pattern. Specifically, we find that socioeconomic status and geography account for only a small share of the transmission. Combined with the absence of a dietary response following a parent’s unexpected lifestyle-related death, these findings underscore the importance of early-life influences and habit formation.

Keywords: health behaviors, intergenerational mobility, diet.

JEL codes: D12, I12, J62.

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1 Introduction

Parents shape many aspects of their children’s lives, from education and career choices to broader measures of well-being. While much of the economics literature has focused on the intergenerational transmission of income and wealth, a growing body of research examines the transmission of health, recognizing its fundamental role in overall well-being (see, e.g., [Mazumder, 2024](#)). Diet is a key determinant of health, and, in light of the ongoing obesity epidemic and the rising prevalence of lifestyle-related diseases, understanding how dietary habits are passed across generations is crucial.¹

This paper provides novel evidence on the intergenerational transmission of dietary choices from parents to children by exploiting unique grocery transaction records linked to administrative data. Specifically, we document a strong intergenerational persistence of dietary choices that exceeds income persistence across all measures we consider, and we further investigate heterogeneities and mechanisms, showing how socioeconomic status, location, family background, and childhood habit formation shape dietary transmission. We leverage customer-linked spending by product categories from 1.7 billion shop visits between the first quarter of 2019 and the second quarter of 2021 at the largest Swiss retailer.^{2,3} We enrich the consumption data with family linkages and individual socio-demographic information from the Federal Statistical Office,

¹A variety of health conditions have been linked to inadequate diet, accounting for 18% of all high-income North American deaths ([Afshin et al., 2019](#)).

²Switzerland provides an insightful case to study dietary patterns, as almost everyone has sufficient access to healthy food. It has a high density of grocery stores, the average distance to the nearest store is just 600 meters, and 80% of the population have a store within 2 kilometers (Swiss Federal Statistical Office). In comparison, the median distance to the nearest food store in the United States is 1,450 meters, and only 40% of the population lives less than a mile from the closest store (USDA). In addition, healthy eating is also relatively affordable in Switzerland, where, according to the World Bank, less than 0.1% do not have the financial means to follow a healthy diet. This is the case for 1.5% of households in the United States and 12% in China.

³Our findings are robust if we concentrate our analyses on the pre-COVID-19 period.

allowing us to observe the shopping behavior of 270,957 individuals (12% of the population of interest) and their parents. The main variable of interest and our measure of the healthiness of an individual's diet is the expenditure share of fresh fruit and vegetables relative to total food expenditures. This measure correlates strongly with recommended nutritional guidelines and key micronutrient intakes, making it a reliable criterion for healthy eating patterns.

In the first part of the paper, we document an extensive intergenerational persistence in dietary choices. We estimate a rank-rank slope of 0.250 and find that children whose parents spend one percentage point more on fruit and vegetables have, on average, a 0.247 percentage point higher spending themselves. Further, the children's probability of reaching the top quintile when parents are in the bottom quintile is 11.5%. This is substantially smaller than the probability that children with parents at the top quintile remain at the top of the distribution (31.9%). We then explore heterogeneities and show that parental influence is stronger in rural areas and among children with lower income or education levels, whereas it weakens with increasing distance between children and parents. Children from families with foreign origins also exhibit greater dietary mobility. Taken together, these findings highlight that higher socioeconomic status and exposure to new social or cultural environments tend to foster healthier eating and loosen intergenerational lock-in.

In the second part of the paper, we study a number of underlying mechanisms that may explain the strong intergenerational persistence of dietary choices. Such mechanisms include the transmission of socioeconomic status across generations, location and network effects, unobserved family backgrounds, such as genetic variations in taste and predispositions to diseases, and habit formation during childhood.

First, because highly educated and high-income individuals tend to eat healthier, we examine the extent to which the intergenerational transmission of these socioeconomic variables drives our results. Using a counterfactual analysis that shuts down the parent-to-child transmission of income and education, we find that the transmission of these variables explains only about 10% of the observed dietary persistence. This finding has important implications. Not only does it suggest that income redistribution may have a limited impact on nutritional inequality, but it also indicates that income and diet are transmitted through different mechanisms.

Second, we examine location preferences, recognizing that families in urban versus rural areas may share local food environments and store access, and that children often live in locations similar to those of their parents. Yet, we find that only 6% of the observed dietary persistence is due to children living in similar spatial environments as their parents.

Third, we analyze whether information about a predisposition to lifestyle-related diseases induces family members to change their diet. Along this line, [Fadlon and Nielsen \(2019\)](#) find significant changes in consumption of preventive care and other health-related behaviors in response to health shocks of a family member. Using death records for the years 2016-2021 together with grocery store data for the same period, we estimate the effect of an unexpected death of a parent from lifestyle-related diseases on the share of fruit and vegetable consumption using a staggered difference-in-differences design. While this shock might highlight shared health risks, we detect no significant effect on children's fruit and vegetable share over the two years following the event. However, consistent with [Oster \(2018\)](#), we find a small decline in convenience food purchases, which suggests that some individuals make an effort to avoid unhealthy food.

Another potential channel we consider is the transmission of taste from parents to children. Parents can affect children's taste through genes (nature) as well as through epigenetics and fetal programming (nurture). To test whether the observed persistence is solely driven by nature, we focus on children of divorced parents, allowing us to observe each parent's diet separately. Due to social norms, most of these children grew up with their mothers. Hence, if the persistence were solely nature-driven, we would expect comparable maternal and paternal transmission. However, the stronger maternal influence that we find points to a non-negligible nurture channel.

Finally, we argue that childhood environment and habit formation are key drivers of our findings. The role of early-life environment in shaping health behaviors is well documented in the nutrition literature (see, e.g., [Birch, 1999](#); [Scaglioni et al., 2018](#)). Similarly, in economics, [Adamopoulou et al. \(2024\)](#) show that early-life meat scarcity increases meat consumption later in life. More importantly, they find intergenerational spillovers, as individuals whose mothers experienced meat scarcity consume more meat themselves, providing further evidence of the long-term influence of early-life conditions on dietary habits across generations.

Our findings contribute to the literature on the intergenerational transmission of health behaviors. While economic research has traditionally focused on income ([Chetty et al., 2014](#), [Bratberg et al., 2017](#), [Corak, 2020](#), [Deutscher and Mazumder, 2020](#), [Acciari et al., 2022](#), [Asher et al., 2024](#)), wealth ([Charles and Hurst, 2003](#), [Clark and Cummins, 2015](#), [Adermon et al., 2018](#), [Belloc et al., 2024](#)), and education ([Black et al., 2005](#)) due to their straightforward measurability and data availability, health is a fundamental component of well-being and a major driver of public expenditures. Accordingly, a growing body of literature studies the intergenerational transmission of health outcomes using different measures, such as longevity ([Black et al., 2024](#)), self-reported health status ([Halliday et al., 2020](#)), quality-adjusted life years ([Bencsik et al., 2023](#)), or health care utilization ([Andersen, 2021](#), [Chang et al., 2024](#)).

Our study also contributes to the broader literature on health behaviors. As lifestyle-related chronic diseases account for a growing share of morbidity, mortality, and healthcare spending, understanding the determinants of health behaviors has become increasingly relevant.⁴ Prior research has examined various health-related behaviors, including sleep, physical activity, and alcohol consumption ([Avery et al., 2025](#), [Charness and Gneezy, 2009](#), [Schilbach, 2019](#)). At the same time, several studies document that dietary patterns remain stable even in the face of major life events or policy interventions, suggesting that habits could be an important source of this persistence.⁵ Particularly, [Oster \(2018\)](#) shows that significant health shocks or informational interventions lead to only minor changes in eating habits.

Motivated by the importance of health behaviors, others explore the role of the family in shaping habits such as smoking ([Darden and Gilleskie, 2016](#); [Li and Gilleskie, 2021](#)) and preven-

⁴According to the American Diabetes Association, every fourth healthcare dollar in the United States is spent on people with diabetes, and patients with diabetes generate more than twice as many medical costs as those without the disease.

⁵This literature studies, for instance, migration ([Bronnenberg et al., 2012](#), [Atkin, 2013](#), [2016](#)), family shocks ([Hut, 2020](#), [Hut and Oster, 2022](#)), supermarket openings ([Allcott et al., 2019a](#)), as well as food subsidies ([Hastings et al., 2021](#), [Goldin et al., 2022](#), [Bailey et al., 2024](#)), food labels ([Araya et al., 2022](#), [Barahona et al., 2023](#)), or sin taxes ([Allcott et al., 2019b](#), [Dubois et al., 2020](#), [Dickson et al., 2023](#), [Kiesel et al., 2023](#)).

tive care (Fadlon and Nielsen, 2019). A few studies have specifically explored the transmission of dietary habits across generations (Tosi and Rettaroli, 2022; Adamopoulou et al., 2024). However, challenges remain, including small sample sizes, measurement difficulties, and reliance on self-reported survey data.⁶ We contribute to this literature by constructing an objective, tractable, and transparent measure of diet quality that overcomes many of these limitations. Using high-frequency grocery transaction data covering 12% of the relevant population, we capture expenditure shares on fresh fruit and vegetables, avoiding self-reported biases, arbitrary cutoffs, and subjective weighting schemes. Similarly, scanner data used, for example, in Oster (2018), although rich in granular purchase information, typically lack detailed socioeconomic information and family linkages. By combining grocery transaction records with administrative data, our study overcomes these limitations, allowing us to examine intergenerational transmission while controlling for socioeconomic background, location preferences, and other relevant factors.

The remainder of the paper is structured as follows. Section 2 introduces the data and presents summary statistics. Section 3 explains our measures of intergenerational mobility. Section 4 presents the results on the intergenerational transmission of diet and compares them to income mobility. In Section 5, we explore the potential mechanisms behind these results, and Section 6 concludes.

2 Data

We analyze the intergenerational transmission of diet by combining (i) individual transaction data from the largest Swiss retailer with (ii) administrative data from the Federal Statistical Office. Throughout this paper, we refer to *children* as adult residents for whom we observe at least one parent in the administrative data. They are our population of interest, and we treat their

⁶Tosi and Rettaroli (2022) construct a diet-quality index from food-frequency and calorie-intake information, while Adamopoulou et al. (2024) analyze meat-consumption frequency and an indicator for unbalanced diets based on categorical cutoffs. These measures are informative but relatively coarse proxies for overall diet quality and rely on self-reported survey data.

parents' characteristics as observable covariates. To introduce the data, we refer to individuals in the grocery data as *customers* and those in the administrative data as *residents*.

2.1 Data Sources

Grocery Transaction Data. The consumption data stems from the loyalty program of the largest Swiss grocery retailer. We observe expenditures on 41 product groups for 1.7 billion customer-linked purchases between 2019Q1 and 2021Q2, and customer characteristics include their residence location, age, and household type.⁷ Locations are coded on a grid of 350,000 100×100-meter cells with a mean population of 25 customers. In this program, participants identify themselves at the checkout with their loyalty cards in exchange for exclusive offers and discounts. The program has substantial coverage, tracking expenditures of 2.1 million active users (32% of all residents above legal age), spending on average at least 50 Swiss francs monthly (USD 56 on July 29, 2024), and capturing 79% of the retailer's total sales. Notably, the retailer charges the same prices throughout the country, independent of local purchasing power, wages, and costs, and stores of comparable size generally offer similar goods, except for local products.

We focus on the children's expenditure share of fresh fruit and vegetables relative to total food expenditures over the sample period. This is a suitable measure for a healthy diet because (i) fruit and vegetables are highly correlated with the healthy eating index in [Allcott et al. \(2019a\)](#) (0.57 and 0.41, respectively), (ii) a diet low in fruit or vegetables is among the most frequent reasons for nutrition-related mortality in [Afshin et al. \(2019\)](#), and (iii) our measure correlates strongly with the intake of important micronutrients across age groups.⁸ Furthermore, this

⁷The retailer held a market share of 32.7% in 2020. The major product groups include, among others, *fruit and vegetables*, *meat and fish*, *eggs and dairy*, and *convenience*. The household types are *small households*, *young families*, *established families*, *golden agers*, and *pensioners*.

⁸We compare our data's fruit and vegetable shares to the micro-nutrient intake reported in the 2014-2015 *National Nutrition Survey* (by age group). This survey inquired 2,000 participants between the ages of 18 and 75 about their previous day's diet. The expenditure share on fruit

measure provides a transparent and objective approximation of dietary quality as it does not require any weighting of different nutrients or products.

Administrative Data. We enrich this unique consumption data with administrative records for the entire Swiss population (8.7 million inhabitants in 2020). Pseudo social security numbers allow linking residents across three different administrative datasets. The *Population and Households Statistics* provides socio-demographic characteristics for each resident for the years 2016–2021. This includes, among others, information on gender, age, marital status, residence location, household identifiers, and family linkages. The residence locations are coded on the same 100×100-meter grid as in the grocery transaction data. The *Old-Age and Survivors Insurance* dataset contains annual gross labor market income for every resident for the years 2016 to 2021. We average annual household income for the years 2016–2021 to reduce biases in permanent income from transitory shocks.⁹ We define adjusted household income by dividing by the square root of household size (including children). Finally, the 2010–2021 *Structural Survey* provides information on the highest level of education completed in a household.

2.2 Sample Construction

We restrict our analysis to customers whom we can uniquely match with residents using the common variables of age and location, and further refine the match with information on household type. The detailed steps of this procedure are provided in the Online Appendix. We then assign the transaction data to all adult residents in a household. The procedure links 337,950 children to at least one parent. We restrict the sample to children and parents with average monthly grocery expenditures, adjusted by the square root of household size, between 50 and 1,000 Swiss francs. This is because too-small monthly baskets might not accurately capture

and vegetables has a correlation of 0.4 with the intake of fiber, 0.38 with phosphorus, 0.33 with zinc, 0.22 with Vitamin A, and 0.29 with magnesium.

⁹We compute the total household’s annual income by summing the income of all household members but excluding grown-up children who still live with their parents, as they likely do not contribute to the household’s budget.

overall consumption, while too-large monthly baskets are unlikely to reflect personal use but are likely to come from business customers. We keep households with at most ten members to exclude large cohabiting arrangements and retirement homes. Ultimately, we focus on *children* between the ages of 21 and 70 and parents between the ages of 48 and 97 to avoid too small age groups in our estimation.¹⁰ Further, we construct parental variables as the weighted average of the father’s and mother’s values, using their respective food expenditures as weights. This results in a final sample of 270,957 children.¹¹

2.3 Summary Statistics

[Table 1](#) displays summary statistics for the children’s monthly food expenditures and the share allocated to fruit and vegetables. The average child’s household spends 399 Swiss francs per month (450 USD on July 29, 2024) and allocates 15% of this money to fruit and vegetables. To put this figure in perspective, the *National Nutrition Survey* indicates that only 12% of Swiss households reach the recommended five daily portions of fruit and vegetables. In the cumulative distribution function of our variable, this point aligns with roughly 23% of grocery spending. At the other end of the distribution, about one-third consume no more than a single portion per day. The last two columns of [Table 1](#) compare expenditures in our data to the administrative *Household Budget Survey*, showing that our transaction data covers 65% of the average household grocery expenditures on food and beverages.¹²

Looking at different characteristics, we find that grocery spending is highest among individuals aged 45–54, whereas the share of expenditures allocated to fruit and vegetables ranges

¹⁰Because we detect minor life cycles in diet, we provide all our results conditional on age groups (see [Section 3](#), for details).

¹¹As a robustness check, we increase the size of the final dataset by employing a probabilistic matching procedure based on machine learning techniques. This approach expands the sample by 32%. Using the larger dataset leaves our findings qualitatively unchanged.

¹²This survey continuously selects 3,000 households each year, and participants take notes on their expenditure for an entire month. Note that, as we do not observe beverage expenditures, our actual coverage of food products is higher.

from 15% among younger cohorts to 17% among older ones. This gives a first indication of a potential – albeit small – life cycle in diet. Food expenditures also grow with income and education: the top income quintile spends 458 Swiss francs per month compared to 269 Swiss francs for the bottom quintile. Children living in wealthier and better-educated households also consume relatively more fruit and vegetables, providing evidence of nutritional inequality across different socioeconomic groups, as previously observed in [Allcott et al. \(2019a\)](#). Finally, we observe a larger fruit and vegetable share in urban areas than in suburban or rural areas. One explanation could be that households in sparsely populated areas are more likely to obtain fresh products from a farmer or own a garden. Yet, children in rural areas spend 386 Swiss francs, which is only marginally less than children in urban areas (389 Swiss francs), so we do not expect this to affect our results.

To assess the representativeness of our data, [Table A1](#) in the Appendix presents summary statistics for the 270,957 matched children and compares them to the 2.3 million children in the population who meet the same selection criteria. The average *child* in the final dataset is 43.7 years old and had a total household income of 142,370 Swiss francs. 54% of them are female and 62% are married. Further, 53% hold a tertiary degree, and 90% live in multi-person households. Regarding geographical characteristics, 77% of the children in our sample live in the German-speaking part of Switzerland, 19% in the French-speaking, and 4% in the Italian-speaking region. Comparing these figures with the population reveals larger discrepancies in household size, population density, income, and marital status. This is expected, as it is easier to uniquely match multi-person households and individuals in less densely populated areas. Since household size is positively associated with income, and married individuals are more likely to live in multi-person households, these factors explain the observed differences in income and marital status. As shown in the table, we can correct for these discrepancies by reweighting our sample to match the population distribution of household size and population density, thereby correcting for the mechanical overrepresentation of certain household types.¹³

¹³Reweightings does not materially affect the intergenerational transmission results.

3 Measuring Mobility

In this paper, we rely on four standard measures of mobility that we introduce below (see [Deutscher and Mazumder \(2023\)](#) for an extensive discussion on these measures). We need to consider that the focus is on diet, not income, and the two outcomes potentially exhibit important differences. While at first glance, it might not be clear that higher consumption of fruit and vegetables above a certain threshold yields additional benefits (as it is for income), recall that 88% of the population falls short of the recommended intake, implying that the vast majority remains below this threshold.

In addition to these considerations, both diet and income exhibit systematic fluctuations over the life cycle, which need to be taken into account when studying intergenerational mobility. [Figure 1](#) illustrates life-cycle patterns in diet and income by plotting average income and the share of fruit and vegetable consumption as functions of age. To ensure comparability, both variables are normalized to their respective lifetime means. While both series fluctuate over the life cycle, the variation in diet is much smaller than in income. Income more than doubles between ages 21 and 60 before declining again toward retirement. By contrast, diet follows an s-shaped pattern: young individuals tend to consume relatively few fruits and vegetables, but intake improves by about 30 percent by age 35. It then declines by roughly 10 percent until age 50, after which it increases again.¹⁴

Given the visible, albeit small, life cycle in diet, and since we observe children and parents at the same point in time, we estimate percentile ranks conditional on age as in [Chetty et al. \(2014\)](#) for the positional measures, and we control for age if the measure directly depends on the share of fruit and vegetables. Unless otherwise indicated, we compare a child's diet to the weighted average of their parents' diet, where the weights are proportional to their expenditures.

Rank-Rank Slope. Our first measure of intergenerational mobility is the rank-rank slope (RRS). Let r_{ci} denote child i 's percentile rank (from 1 to 100) and let r_{pi} be the corresponding

¹⁴For both variables, the graph shows the values of the variable at a point in time. Thus, both age and cohort effects could drive these differences. The increase later in life could also partly reflect higher survival rates among individuals with healthier diets.

parent's variable. The percentile ranks are computed within each age category. We fit a rank-rank regression by regressing the children's percentile rank on the parents' percentile rank:

$$r_{ci} = \alpha + \beta r_{pi} + \epsilon_i, \quad (1)$$

where β is the rank-rank slope, which provides a measure of transmission of the parents' position in their generation. A value of $\beta = 0.3$ implies that if you compare two sets of parents one decile apart, their children are expected to be three percentiles apart. A steeper slope reflects a less mobile society. For instance, if each child were in the same percentile as their parents, the slope would be one.

Intergenerational Elasticity. As a second measure, we directly examine the relationship between children's diet and their parents. This measure is similar to the well-established intergenerational elasticity computed by regressing the logarithm of children's income on the logarithm of parents' income. Instead of taking logarithms, we use a quadratic specification, which better fits the data, as shown in the following regression model:

$$s_{ci} = \delta_1 s_{pi} + \delta_2 s_{pi}^2 + x_i' \gamma + v_i, \quad (2)$$

where s_{ci} and s_{pi} are, respectively, the child's and parents' fruit and vegetable share, and x_i contains the child's and parents' ages, along with their squared terms. Since we fit a polynomial regression, the slope changes over s_{pi} . We therefore report the average marginal effects (AME), which we refer to as the *intergenerational elasticity* (IGE), with a slight abuse of terminology.

Conditional Expected Rank. The conditional expected rank (CER) is the expected percentile rank of children having parents at percentile p :

$$CER(p) = \mathbb{E}[r_{ci} | r_{pi} = p]. \quad (3)$$

We focus on the CER at the 25th and 75th percentiles, denoted CER25 and CER75. The CER can be estimated parametrically (from a rank-rank regression) and nonparametrically. We opt

for a middle ground and use a nonparametric local linear regression evaluated at percentile p .

Transition Matrix. Transition matrices break down the children’s and parents’ distribution into groups of equal size. We group children and parents conditional on age into quintiles and compute the conditional probability that a child is in bin p_j given their parents are in bin p_k :

$$TP_{j,k} = Pr(s_{ci} \in p_j | s_{pi} \in p_k). \quad (4)$$

4 Intergenerational Persistence Results

This section presents results on the persistence of dietary habits across generations. [Table 2](#) reports coefficients and standard errors computed using 1,000 nonparametric bootstrap replications. Further, to assess the magnitude of the persistence of dietary choices, we compare the findings to intergenerational mobility in income.

4.1 Dietary Mobility

Rank-Rank Regression. Panel (a) of [Table 2](#) shows an estimated rank-rank slope of 0.250, indicating that a 10-percentile increase in the parental dietary rank is associated with a 2.5-percentile increase in the child’s rank. To put this into perspective, it would take approximately 3.16 generations to close the gap between two families at the 10th and the 90th percentiles.¹⁵

To show that conditioning the percentile ranks on age addresses life-cycle fluctuations, in the Online Appendix, we present rank-rank regression results using conditional and unconditional percentile ranks, allowing the intercept and the slope to change over the life cycle by saturating the model in children’s age. While the slope is almost identical across both specifications, the intercept largely depends on the specification of the percentile ranks, and when using unconditional percentile ranks, the intercept captures the life cycle observed in [Figure 1](#). This observation suggests that conditional percentile ranks are a better measure of dietary mobility

¹⁵The number of generations N to close the gap of $\Delta_{10,90} = 80$ percentile ranks between the first and ninth decile solves $\beta^N \Delta_{10,90} = 1$, such that $N = \frac{\log(1/\Delta_{10,90})}{\log(\beta)}$.

than their unconditional counterparts. The rank-rank slope is large and roughly constant in early adulthood at around 0.27, showing that dietary habits acquired at an early age carry on far into adulthood. The slope starts declining at around age 45, which could be explained by habit adaptation, taking several periods to form. Yet, the relationship remains remarkably sizable well into later life.¹⁶

Intergenerational Elasticity. Panel (b) of [Table 2](#) reports the estimated average marginal effect. A one percentage point increase in parents' fruit and vegetable consumption is, on average, associated with a 0.247 percentage point increase in their children's consumption. Although not shown in the table, the slope declines at higher levels of parental consumption, indicating diminishing returns. This pattern suggests that targeted policy interventions aimed at families with particularly poor diets may yield greater improvements in children's nutritional behavior.

Conditional Expected Ranks. Panel (c) in [Table 2](#) presents the estimated conditional expected rank. Children whose parents are at the 25th percentile of fruit and vegetable consumption reach, on average, the 45th percentile in their own generation. In contrast, those with parents at the 75th percentile are expected to reach the 56th percentile. These results point to substantial intergenerational persistence in diet, yet also reveal meaningful reversion to the mean.

Transition Matrix. [Figure 2](#) shows the estimated transition matrix with the corresponding confidence intervals. We include selected key results of the transition matrix in [Table 2](#), Panel (d). Without intergenerational persistence of diet across generations, we would observe 20% of children in each cell. The estimated transition probabilities reveal a strong persistence in diet between generations, as children are most likely to be in the same quintile as their parents. Focusing on the tails of the parents' distribution, we find that 31.3% of children whose parents spend the least on fruit and vegetables also fall at the bottom of the distribution (corresponding

¹⁶Estimates for older age groups are noisier due to smaller sample sizes. Note that, as before, both age and cohort effects could explain these differences.

to a Q1Q1 transition), while only 11.5% move up to the top (Q1Q5). If, on the other hand, an individual's parents are in the top 20% of fruit and vegetable consumers, their children are also most likely to be in the fifth quintile (Q5Q5). These particularly strong results in the “extreme” transition probabilities provide evidence that the so-called cycles of poverty and privilege are pronounced. At the same time, mobility appears larger around the center of the distribution.

4.2 Comparison to Income Mobility

To put the magnitude of our findings into perspective, we compare them to intergenerational mobility in income. More specifically, we focus on the relationship between children's income and their fathers' income. To deal with life-cycle variation in income, we follow the previous literature and focus on a subgroup of children and fathers with stable income (see, among others, [Chetty et al., 2014](#), [Corak, 2020](#), or [Acciari et al., 2022](#)). We generate a new data set for children between 30 and 40 with fathers between 50 and 62. This restriction ensures that most children are already participating in the labor market and fathers are not yet retired. Further, we average income over the years 2016–2021 to smooth out transitory fluctuations. We estimate the same measures for intergenerational income mobility that we use for diet, again conditioning on age, and present the results in [Table 2](#). We estimate a rank-rank slope of 0.145 and an intergenerational elasticity of 0.113.¹⁷ The conditional expected ranks at the 25th and 75th percentiles are 47.48 and 53.98. Also, more than one in four children with fathers at the bottom quintile stay at the bottom, and 14.1% move up to the top.¹⁸ Comparing our estimated mobility measures between diet and income in [Table 2](#), we find that intergenerational transmission is consistently stronger for dietary behavior across all metrics. [Figure 3a](#) underscores this difference graphically: the slope of the rank-rank regression for diet (darker line) is substantially steeper than for income

¹⁷We estimate the intergenerational elasticity using a classical log-log specification, however, including also a quadratic term.

¹⁸Different sample selection procedures and income definitions lead to comparable findings. Furthermore, our estimates on income mobility in Switzerland are in the range of those in [Chuard and Grassi \(2020\)](#), who measure the parental income as the average of the father's and mother's income when the child is between 15 and 20 years old.

(lighter line), with the dots representing the average child percentile rank at each parental percentile rank.

Such pronounced persistence in dietary choices – an important health behavior – may help explain broader intergenerational persistence in health status. At the same time, this strong relationship suggests that the development of dietary habits during childhood could be a persistent channel through which parents influence their children, potentially to a greater extent than their influence on economic outcomes. It is worth noting that income is particularly mobile in Switzerland, so the persistence of diet relative to income may differ in other countries.¹⁹

4.3 Other Dietary Measures

To complement our main analysis based on fruit and vegetable consumption, we extend our investigation to additional dietary dimensions. These include expenditure shares on meat and fish, eggs and dairy, and convenience food (i.e., ready-made or processed items requiring minimal preparation) as well as binary indicators for being vegetarian, a heavy meat consumer, or a high user of convenience food. While these categories are less directly interpretable in terms of healthfulness, given their broader nutritional profiles, they offer a richer view of intergenerational dietary transmission. The results, reported in the Online Appendix, reveal substantial persistence across all categories. Although the strength of transmission for these additional measures is somewhat lower than for fruit and vegetables, it still exceeds that of income, underscoring the deep-rooted nature of dietary behaviors across generations. We focus on fruit and vegetable consumption in the main analysis because it reflects a homogeneous and widely endorsed component of a healthy diet, in contrast to the more heterogeneous nature of other food categories.

¹⁹Previous studies report rank-rank income slopes of 0.34 for the U.S. ([Chetty et al., 2014](#)), 0.24 for Canada ([Corak, 2020](#)), 0.22 for Sweden and Norway ([Bratberg et al., 2017](#)), 0.25 for Italy ([Acciari et al., 2022](#)), and 0.21 for Australia ([Deutscher and Mazumder, 2020](#)).

4.4 Heterogeneities

Heterogeneities in the persistence of dietary choices across socioeconomic dimensions may allow some individuals to adopt healthier diets while leaving others more constrained. This section explores differences by income, education, urbanization, distance to parents, and national origin, and sets the stage for the analysis of potential mechanisms, which we examine in detail in the following section. To correct for a possible mechanical result that children belonging to an unhealthy group have a higher chance of surpassing their parents, we use percentile ranks based on the entire sample but reweight observations within each group so that the parents' distribution mirrors that of the full sample.²⁰

Table 3 shows the results for the different subgroups. The second column reports the p -value from a Wald test of equal rank-rank slopes across the subgroups. First, Panel (a) shows the results for the three education levels: primary, secondary, and tertiary. The rank-rank slopes lie around 0.23 in all groups and are not statistically different from each other. This suggests that parental transmission of diet is unaffected by children's educational attainment. Instead, the intercept increases with education, indicating that more highly educated children consume more fruit and vegetables. Therefore, education allows children to break out of unhealthy dietary habits, not through a change in the transmission of these habits but through the simple fact that higher-educated households systematically follow a healthier diet, independent of their parents. Multiple reasons may explain this observation. For example, highly educated individuals may have a more profound nutritional knowledge, a better assimilation of dietary information, or higher patience.

Second, Panel (b) digs into differences between income groups.²¹ As shown, the rank-rank slope and intergenerational elasticity monotonically decrease as children's income increases. For the poorest children, we find a rank-rank slope of 0.279 compared to 0.208 for the richest

²⁰For instance, the reweighting procedure gives equal weights to all percentiles in the rank-rank regression and the conditional expected rank. For an extensive discussion of weighting approaches in these settings, see [Deutscher and Mazumder \(2023\)](#).

²¹To account for the life cycle in income, we condition income quartiles on age and focus on working-age children (25-64). The results are unaffected if we use all observations.

individuals. These differences are also statistically significant, suggesting that percentile ranks are more persistent over generations among low-income children. [Figure 3b](#) shows the rank-rank regression lines for all four income quartiles. The differences in intercepts and slopes suggest that low-earning children are less successful at breaking unhealthy childhood habits and at maintaining beneficial ones. For instance, a high-earning child with parents at the 10th percentile has the same expected rank as a low-earning child whose parents are at the 70th percentile.

These heterogeneities are also visible across geographical characteristics. Panel (c) shows that mobility is highest in urban areas and lowest in rural areas. The transition probabilities show that children living in urban areas have an outstanding likelihood of moving up in the distribution. Strikingly, a child born to parents in the first quintile of the distribution who lives in an urban area is only 2.5 percentage points more likely to find himself at the bottom of the distribution than at the top. For children living in suburban (rural) areas, this difference is over 20 (27) percentage points.

Panel (d) analyzes the role of the distance between children and parents. We observe that nutritional persistence remains high even if children live far away from their parents. However, the further the children move away from their parents, the lower the persistence. This result is not surprising, as living away from one's family is often associated with moving away from one's childhood environment. However, it is striking that children are eight percentage points less likely to be trapped at the bottom if they live far away. This finding suggests that new social networks and environments might play a decisive role in breaking old habits and is consistent with previous findings on diminishing social interactions and responses to family-related shocks with increasing distance (see, e.g., [Fadlon and Nielsen, 2019](#) and [Büchel et al., 2020](#)).

An additional analysis indicates that while shared store preferences, shopping frequency, and the degree of urbanization influence the intergenerational persistence of dietary habits, they have limited explanatory power for the distance-based heterogeneity. Dropping children who shop at the same store decreases the distance gradient only slightly. Similarly, the distance gradient remains substantial across rural, suburban, and urban subsamples – despite differences in store density. Further, we find that shopping frequency has a negligible effect on dietary

composition. These findings suggest that shared grocery habits and store preferences do not fully explain the observed heterogeneity, pointing instead toward deeper behavioral mechanisms and network effects.

Finally, in Panel (e), we examine heterogeneity by nationality, using parental nationality to distinguish between children of Swiss and foreign origin. We find meaningful differences: dietary habits are more strongly transmitted in Swiss households, while those of foreign origin exhibit greater dietary mobility. This result is not surprising, as individuals with foreign origins are more likely to be exposed to peers whose backgrounds differ from those of their own families, leading to more diversified social networks. This aligns with the idea that migrants may be more likely to adapt their eating habits to new environments, leading to weaker intergenerational persistence (see [Atkin, 2013, 2016](#)).

5 Mechanisms

In the previous subsection, we documented a strong intergenerational persistence in dietary habits. At the same time, we observe substantial heterogeneity in this persistence, raising important questions about the underlying mechanisms. We now consider a number of potential channels, including the transmission of socioeconomic status across generations, location and network effects, unobserved family background (such as genetic variations in taste and predispositions to diseases), and habit formation during childhood.

5.1 Socioeconomic Status

We have shown that mobility is significantly higher among better-earning and higher-educated children, who also tend to follow healthier diets. Given the well-documented intergenerational persistence of income, a natural question arises: To what extent does the transmission of socioeconomic status drive dietary persistence? This subsection isolates and quantifies the component of intergenerational transmission in diet that cannot be attributed to the transmission of income and education. To isolate this channel, we compute counterfactual distributions in the spirit of [Chernozhukov et al. \(2013\)](#). More precisely, we are interested in estimating the conditional

distribution of the children's diet (conditional on their parents' diet) that we would observe if their income and education were independent of their parents' socioeconomic variables.²² This counterfactual distribution is identified by combining the cumulative distribution function (cdf) with an alternative covariate distribution. Then, from this counterfactual distribution, we can easily compute a counterfactual transition matrix.

More formally, let $F_{s_c|s_p}$ be the cdf of children's diet s_c conditional on the parents' diet s_p and the ages of children and parents, a_c and a_p . All the distributions outlined in the section are conditional on the age variables; however, we omit the dependency for notational ease unless needed for clarity. Let x_c denote a vector containing the children's income and education, and let x_p contain the corresponding parental variables. The main object of interest is the counterfactual distribution of the children's diet that we would observe if we change the covariate distribution $F_{x_c|s_p,x_p}(x_c|s_p,x_p)$ to a different distribution $F_{x'_c|s_p,x_p}(x_c|s_p,x_p)$. We denote this counterfactual distribution $F_{s_c|s_p}\langle x_c|x'_c\rangle(s_c|s_p)$. Starting from the cdf of the children's diet conditional on (s_p, x_p, x_c) , we can attain $F_{s_c|s_p,x_p}\langle x_c|x'_c\rangle(s_c|s_p, x_p)$ by integrating the conditional cdf over the alternative covariate distribution:

$$F_{s_c|s_p,x_p}\langle x_c|x'_c\rangle(s_c|s_p, x_p) = \int_{X'_c} F_{s_c|s_p,x_c,x_p}(s_c|s_p, x_c, x_p) dF_{x'_c|s_p,x_p}(x_c|s_p, x_p), \quad (5)$$

²²A least squares regression of children's diet on parents' diet, controlling for socioeconomic variables, does not disentangle this effect for several reasons. First, we need to model the distribution of children's diets to analyze directional mobility. Second, a least squares regression would fix a socioeconomic variable, whereas we want to consider a specific change in the covariate distribution. Third, comparing regressions that control for income and education with a regression without these controls provides meaningful results only under the strong assumptions of the correct specification. As we show in [Table 3](#), diet transmission is heterogeneous across socioeconomic status, violating this assumption. While it would be possible to estimate a more flexible model that includes interactions between s_p and socioeconomic variables, such a model would become extremely tedious to interpret. Instead, by estimating counterfactuals, results remain straightforward to interpret.

where X_j denotes the support of the covariates x_j for $j = \{c, p\}$ conditional on the other variables. Then, integrating Equation (5) over the distribution of the parents' covariates yields the desired result:

$$F_{s_c|s_p} \langle x_c | x'_c \rangle (s_c | s_p) = \int_{X_p} F_{s_c|s_p, x_p} \langle x_c | x'_c \rangle (s_c | s_p, x_p) dF_{x_p|s_p}(x_p | s_p). \quad (6)$$

In the counterfactual scenario we consider, we close the pathway from parents' diet to children's diet operating through the intergenerational transmission of education and income. Consequently, the counterfactual distribution of children's income and education should be independent of parental socioeconomic variables. We further assume that parents' age and diet do not affect children's characteristics. Thus, the counterfactual covariate distribution is the conditional distribution of x_c given a_c , where children's age in the conditioning set captures life-cycle changes in income as well as differences in the distribution of education across cohorts.

The estimation follows the plug-in approach. We obtain an estimate of the conditional cdf $F_{s_c|s_p, x_c, x_p}$ by inverting the estimated conditional quantile function. We estimate the conditional quantile function by fitting a flexible quantile regression model for the set of quantiles $\{0.005, 0.015, \dots, 0.995\}$. The regressions include a second-order polynomial of the parents' diet. Further, we include age and education dummies as well as household income (and its square) interacted with age and a dummy for age ≥ 65 for both parents and children. This last term allows income to have a different effect over the life cycle, which is discontinuous after reaching retirement age. All variables are interacted with a second-order polynomial of the parents' diet. We estimate $F_{x_c|a_c}$ using the empirical distribution function.

For this analysis, we restrict the sample to the 135,213 children for whom we observe both their own and their parents' education. The validity of this approach relies on the correct specification of the conditional quantile function. While we fit a flexible model, we re-estimate the baseline transition probabilities in this smaller sample using the same quantile model to further ensure a meaningful comparison.

Figure 4a shows the estimated transition probabilities with the corresponding bootstrap 95% confidence intervals. The left matrix displays the transition probabilities estimated with the procedure described above; however, it uses the original covariate distribution. These

results are statistically indistinguishable from the nonparametric results for the entire sample in [Figure 2](#). The right matrix shows the counterfactual transition probabilities. While the transition matrices are remarkably similar, mobility is statistically significantly higher in the counterfactual scenario, particularly at the extremes of the distribution. For example, the Q1Q1 and Q5Q5 probabilities decrease, and the Q1Q5 probability increases. Consider the Q5Q5 cell: in the original transition matrix, individuals whose parents are in the fifth quintile are 11.6 ($= 31.6 - 20.0$) percentage points more likely to be in the fifth quintile than under no intergenerational transmission of diet. We refer to this as an excess probability. In the counterfactual scenario where we close the channel going through income and education, this number declines to 10.6 ($= 30.6 - 20.0$) percentage points. This implies that the intergenerational transmission of income and education explains less than 9% of this excess probability. A similar calculation indicates that around 10% of the excess probability of remaining trapped at the bottom of the distribution is explained by income and education transmission.

To summarize these transition matrices with a single number, we compute the normalized anti-diagonal trace, following [Jäntti and Jenkins \(2015\)](#). The normalization consists of subtracting the anti-diagonal trace of a completely mobile society. For the transition matrix on the left, we find a normalized anti-diagonal trace of 28.4. For the counterfactual matrix, this statistic equals 25.9, suggesting that income and education transmission drive only 8.8% of the results.²³

These results suggest that only a small share of the intergenerational persistence in diet can be explained by the transmission of income and education. This finding is surprising and carries important implications. Even if income and education were fully mobile across generations, strong dietary persistence would remain, implying that policies focused solely on income redistribution or financial assistance may have a limited impact on reducing nutritional inequality. This conclusion aligns with evidence showing that monetary incentives have had only modest success in promoting healthier food choices among SNAP recipients (see, for

²³Starting from the counterfactual distribution, we can compute all mobility measures, and we find that the results are consistent across the different metrics. Further, the results are robust if we drop children and parents who either have zero income or are retired.

example, [Verghese et al. \(2019\)](#) and references therein). More broadly, the findings point to the existence of distinct transmission channels for diet and income. Since well-being spans multiple dimensions that are inherited through different mechanisms, focusing exclusively on income mobility provides an incomplete picture. A deeper understanding of these separate pathways is therefore essential for designing more effective policies to address disparities in overall well-being.

5.2 Current Location

Besides socioeconomic characteristics, the transmission of location preferences might partly explain our results. Yet, these variables are more difficult to measure than income or education, and more importantly, it is unclear which characteristics of a location are meaningful in determining diet. In this analysis, we use population density as a broad measure of location characteristics, which tends to persist across generations.²⁴ Because individuals in urban areas tend to follow healthier diets, the intergenerational transmission of location may partially contribute to the persistence of dietary habits.

To assess the contribution of location to dietary persistence, we repeat the exercise used for income and education, but now remove the intergenerational link in location, measured by degree of urbanization. Specifically, we construct a counterfactual in which an individual's probability of living in an urban, suburban, or rural area is independent of their parents' location.

We again fit a flexible quantile regression model where we interact parents' diet with dummies for the degree of urbanization. [Figure 4b](#) displays the original and the counterfactual transition matrices.²⁵ Comparing the normalized anti-diagonal traces of the two matrices, we

²⁴Children who grew up in rural (urban) areas are more likely to live in rural (urban) areas later in life. For instance, 55% of individuals in our sample whose parents live in rural areas also live in a rural area, while only 9% of them reside in an urban area. Similarly, 51% of individuals whose parents live in urban areas also live in an urban area, while only 12% of them reside in a rural area.

²⁵As before, we recompute the original matrix using the same flexible model. The marginal differences in the results compared to the left matrix of [Figure 4a](#) are due to the slightly different

conclude that only 6.0% of the dietary transmission can be explained by children living in similar spatial environments as their parents.

This analysis suggests that, although location is an important determinant of diet, the intergenerational transmission of urbanization contributes little to dietary persistence. These results align with previous papers documenting limited adaptations in diets in response to changes in spatial environments (see, for example, [Atkin, 2013, 2016](#), or [Allcott et al., 2019a](#)).

5.3 Dietary Response to Health Information Shocks

Diet is a key health behavior and may also be shaped by health-related goals. For instance, a revealed predisposition for a lifestyle-related disease could motivate family members to consciously adjust their eating behaviors. To assess the relevance of this channel, we examine how the death of a parent from a lifestyle-related disease affects their children's diet. Such events may serve as informative health shocks, especially given that individuals with a high genetic risk for heart disease nearly double their risk of stroke or heart attack, while a healthy lifestyle can cut this risk by half ([Khera et al., 2016](#)).

For this analysis, we combine grocery store data and administrative vital statistics covering the years 2016 to 2021. The vital statistics include information on all deaths, including underlying conditions. We use a staggered difference-in-differences design where we compare the diet of children whose parents die from a lifestyle-related disease (stroke and heart attack) to that of children who face the same shock in later years. To capture an unexpected shock, we focus on deaths without preexisting conditions. We apply the estimator proposed by [Callaway and Sant'Anna \(2021\)](#) and present the results in an event-study plot. [Figure 5](#) shows no change in fruit and vegetable intake for up to two years after the shock. The coefficients are small and not statistically significant, suggesting that individuals might not perceive this shock as informative about their own risk for lifestyle diseases or simply do not respond to this information.²⁶

samples.

²⁶Although not reported in the paper, we find no meaningful heterogeneity in the effect when examining differences by the gender of the child or deceased, age at death, or children's socioeconomic characteristics.

We complement this analysis by examining the impact of the shock on expenditure shares of meat and fish, convenience food, and eggs and dairy. As shown in [Table 4](#), the estimated effects are generally close to zero. The only statistically significant effect is a small reduction in spending on convenience food: following the death of a parent, individuals reduce their expenditure share on convenience food by 0.285 percentage points (2.3%), on average.

These findings are consistent with [Oster \(2018\)](#) and [Hut and Oster \(2022\)](#), who find only minimal dietary improvements following a diabetes diagnosis, but contrast with [Fadlon and Nielsen \(2019\)](#), who document sizable behavioral responses to health shocks within families. In their setting, two mechanisms help explain why individuals change health behaviors: such shocks provide new information about personal risk and make these risks more salient. Our null results suggest that these mechanisms may be less effective in the case of diet. One reason could be that diet – unlike actions such as taking preventive medication or undergoing screenings – involves higher daily effort, sustained self-control, and delayed rewards. That is, even when individuals receive salient signals about their health risks, adjusting dietary choices may be too costly to act upon. Still, the modest effect on convenience food may reflect some limited effort to avoid unhealthy options, as observed in [Oster \(2018\)](#).

One limitation of our analysis is that we consider only information shocks, and other events, such as diagnoses, could have a more substantial effect on diet, as the affected person might receive or seek nutritional advice from a physician and pass it on to relatives. Nonetheless, [Hut and Oster \(2022\)](#) find only minimal and non-lasting effects of such diagnoses.

5.4 Taste

It is well understood that our taste influences dietary choices. Our families affect our taste both genetically (nature) and through environmentally mediated biological processes such as epigenetics and fetal programming (nurture). These two channels provide explanations of how persistence in taste might drive our results.

A child's genotype, determined at conception and fixed throughout life, influences how sensory signals, such as those from taste receptors, are perceived and processed. Variations in taste and olfactory receptor genes shape sensitivities for certain flavors and foods, thereby

affecting preferences and intake.²⁷ At the same time, genes do not act in isolation; they interact with environmental factors. For example, parental diets may reinforce a child's genetically influenced taste preferences. Fetal programming suggests that maternal diet during pregnancy can have lasting effects on a child's metabolic health and dietary preferences. For instance, suboptimal maternal nutrition can induce fetal adaptations that may increase the risk of specific eating behaviors or metabolic disorders (see, e.g., [Barker, 1990](#); [Almond and Currie, 2011](#)). Similarly, epigenetic mechanisms (i.e., environmentally induced changes in gene expression) are not only a key biological pathway through which fetal programming can occur, but may also provide an additional biological channel for dietary persistence across generations. These modifications can influence taste, metabolic efficiency, or predispositions toward specific foods.

Evidence supports the relevance of these pathways, showing that food intake, even during pregnancy and breastfeeding, can influence a child's taste preferences. For example, reducing sodium and sugar consumption sharpens the perception of saltiness and sweetness ([Wise et al., 2016](#)). Similarly, infants show greater initial acceptance of fruit and vegetables when their mothers consume them regularly during pregnancy ([Mennella et al., 2001](#); [Forestell, 2024](#)) and breastfeeding ([Forestell and Mennella, 2007](#)).

To assess the importance of genetic variation in taste, we analyze matrilineal and patrilineal transmission of diet for the subsample of children with divorced parents who never remarried and live alone, allowing us to observe each parent's diet separately. Due to social norms, most of these children grew up with their mothers.²⁸ Hence, if the dietary transmission were

²⁷Evidence is particularly rich for genetic influences on the perception of bitter ([Mennella et al., 2005](#), [Gervis et al., 2023](#)), sweet ([Mennella et al., 2005](#), [Mennella et al., 2016](#), [Søberg et al., 2017](#)), and alcohol ([Allen et al., 2014](#)), as well as for associations with the consumption of specific foods such as fruit, cheese, fish, and tea ([Cole et al., 2020](#)).

²⁸We focus on divorced parents who did not remarry to avoid contamination from a new partner. According to the *Federal Statistical Office*, around 80% of children under 14 with separated parents lived with their mother in the year 2000. More recent data from the *Federal Department of Home Affairs* (2022) confirm that children continue to spend the majority of nights with their mother far more often than with their father.

only due to the genotype-driven differences in taste (nature), we should see no difference between transmission from mother and father. In contrast, a stronger link to the mother’s diet indicates that the nurture channel is non-negligible.²⁹ The results in [Table 5](#) show that the intergenerational link is substantially stronger for mothers than for fathers. This relationship changes only slightly with the child’s age at divorce.

Taken together, the findings from the literature and our empirical results suggest that parents influence children’s dietary preferences not only through genetic transmission but also through a range of environmental and behavioral channels.

5.5 Habits

In the previous subsections, we showed that socioeconomic variables have only limited explanatory power for the intergenerational transmission of diet, indicating that the mechanisms by which diet and socioeconomic status are passed across generations may differ fundamentally. At the same time, the data shows substantial dietary persistence later in life, and we observe minimal (if any) changes in diet following an information shock about disease predisposition. Taken together, with the much stronger matrilineal resemblance among children of divorcees, these findings point to habits as a central driver. This would imply that diet, like other health-related behaviors such as smoking, exercise, and sleep, is shaped by habits that may take root early in childhood and are influenced by parents.

These habits likely reflect a wide range of *nurture-based* components, including environmentally mediated biological processes (see [Section 5.4](#)), diet-related knowledge, and skills that parents pass on to their children. This is consistent with longstanding evidence from the psychology and nutrition literature showing that the family plays a central role in shaping children’s food preferences (e.g., [Birch, 1999](#); [Scaglioni et al., 2018](#)). Building on these results, [Atkin \(2013\)](#) provides a habit formation model to explain how local food cultures emerge and persist over time. Similarly, [Fadlon and Nielsen \(2019\)](#) highlight the importance of the family in shaping health behaviors. Recent evidence by [Adamopoulou et al. \(2024\)](#) further supports the

²⁹There is no genetic reason for stronger maternal than paternal dietary transmission, as taste receptor genes are not located on the X or Y chromosomes ([Bachmanov and Beauchamp, 2007](#)).

behavioral channel, showing that parents' eating patterns significantly shape their children's dietary choices. These findings resonate with our own and underscore the importance of viewing diet as a learned behavior shaped by early and repeated exposure.

While parental diet is likely a major determinant of a child's *habit stock*, many other factors, including childhood networks and location, may also play a role (see, for example, [Story et al. \(2008\)](#) for an overview). Understanding the determinants of this habit stock and distinguishing them from genetic influences is crucial for informing effective policy. If habits are formed early and remain stable over time, interventions targeting children and their environments may yield larger effects than those aimed solely at adults.

Unfortunately, we do not observe habits directly. Instead, we observe realized food purchases, which we interpret as the outcome of an underlying and unobserved habit stock. In the Online Appendix, we introduce a simple framework on habit formation (see, for example, [Campbell and Cochrane \(1999\)](#), [Fuhrer \(2000\)](#), and [Carroll et al. \(2000\)](#) for some early work on habit formation models). We model the persistence of diet between generations as the result of a habit stock built during childhood and adjusting over a lifetime. The results of the model point to the slow alteration of habits as an important driver of our results.

6 Conclusion

While most research on intergenerational mobility has focused on outcomes such as income and education, this paper turns to an important but often overlooked health behavior: diet. As a key determinant of health, it may serve as a crucial channel through which health and well-being are transmitted across generations. At the same time, the detrimental consequences of poor diets impose a significant social and economic burden. This paper sheds light on the intergenerational transmission of dietary habits by combining unique supermarket transaction data with administrative records containing family linkages. We provide novel evidence that family background is a strong predictor of long-run eating patterns, suggesting that the diet consumed at one's parents' dinner table plays a lasting role in shaping nutritional preferences throughout life.

Our results show that the intergenerational transmission of diet varies across observable covariates. Higher-educated and better-earning children generally eat better, independent of their parents. While the transmission mechanism (in terms of the rank-rank slope) does not vary by education level, it grows significantly weaker as income rises. Hence, low-income individuals are particularly vulnerable to getting stuck in a cycle of unhealthy diets. Moreover, upward mobility is larger among children living in urban areas, and the transmission becomes weaker as the distance between children and their parents increases, suggesting that distancing oneself from the childhood environment may help break unhealthy patterns.

We explore potential drivers of dietary persistence, including the intergenerational transmission of income and education, as well as family background. Isolating the part of dietary transmission going through education and income, we show that the transmission of these socioeconomic variables is responsible for only 10% of the intergenerational persistence of diet. A similar exercise for location preferences reveals that they explain around 6% of the persistence. Further, we find that the unexpected death of a parent due to a lifestyle-related disease does not affect diet, suggesting that dietary change may be too costly for individuals, even when faced with new information about health risks. Similarly, the stronger persistence we observe between mothers and children than between fathers and children among children of divorcees further underscores the importance of the nurture component of intergenerational transmission. Although other unobserved factors likely influence eating habits, our results point to a substantial direct effect of childhood diet and suggest that habit formation is a key mechanism – implying not only that the apple does not fall far from the tree, but also that it remains close to where it lands.

These findings have important implications for public health and policy. Recognizing the strong influence of family on dietary choices can help guide more targeted interventions aimed at fostering healthier eating habits. By identifying the origins of unhealthy eating patterns and understanding how they are transmitted across generations, policymakers and healthcare professionals can design strategies to address the growing burden of diet-related diseases more effectively. Our results suggest that lump-sum transfers or SNAP benefits are unlikely to shift entrenched dietary behaviors. Instead, interventions that directly target children's diets, while

habits are still forming, may prove more effective and cost-efficient. Examples include healthy school lunch programs, nutritional education, and informational campaigns in schools and medical settings. Future work should focus on assessing the effectiveness of such interventions.

Finally, this paper shows that even in a high-income country with relatively uniform access to healthy food and a dense grocery-retail network, diet is highly persistent across generations. Building on this baseline, researchers should test how variation in economic disparities and access to healthy food shape the intergenerational transmission of diet.

References

- ACCIARI, P., A. POLO, AND G. L. VIOLANTE (2022): “And Yet It Moves: Intergenerational Mobility in Italy,” *American Economic Journal: Applied Economics*, 14, 118–163.
- ADAMOPOULOU, E., E. OLIVIERI, AND E. TRIVIZA (2024): “Eating habits, food consumption, and health: The role of early life experiences,” *European Economic Review*, 166, 104743.
- ADERMON, A., M. LINDAHL, AND D. WALDENSTRÖM (2018): “Intergenerational Wealth Mobility and the Role of Inheritance: Evidence from Multiple Generations,” *The Economic Journal*, 128, F482–F513.
- AFSHIN ET AL. (2019): “Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017,” *The Lancet*, 393, 1958–1972.
- ALLCOTT, H., R. DIAMOND, J.-P. DUBÉ, J. HANDBURY, I. RAHKOVSKY, AND M. SCHNELL (2019a): “Food Deserts and the Causes of Nutritional Inequality,” *The Quarterly Journal of Economics*, 134, 1793–1844.
- ALLCOTT, H., B. B. LOCKWOOD, AND D. TAUBINSKY (2019b): “Should We Tax Sugar-Sweetened Beverages? An Overview of Theory and Evidence,” *Journal of Economic Perspectives*, 33, 202–227.
- ALLEN, A. L., J. E. MCGEARY, AND J. E. HAYES (2014): “Polymorphisms in TRPV and TAS2Rs Associate with Sensations from Sampled Ethanol,” *Alcoholism: Clinical and Experimental Research*, 38, 2550–2560.
- ALMOND, D. AND J. CURRIE (2011): “Killing Me Softly: The Fetal Origins Hypothesis,” *Journal*

of Economic Perspectives, 25, 153–172.

ANDERSEN, C. (2021): “Intergenerational health mobility: Evidence from Danish registers,” *Health Economics*, 30, 3186–3202.

ARAYA, S., A. ELBERG, C. NOTON, AND D. SCHWARTZ (2022): “Identifying Food Labeling Effects on Consumer Behavior,” *Marketing Science*, 41, 982–1003.

ASHER, S., P. NOVOSAD, AND C. RAFKIN (2024): “Intergenerational Mobility in India: New Measures and Estimates across Time and Social Groups,” *American Economic Journal: Applied Economics*, 16, 66–98.

ATKIN, D. (2013): “Trade, Tastes, and Nutrition in India,” *American Economic Review*, 103, 1629–1663.

——— (2016): “The Caloric Costs of Culture: Evidence from Indian Migrants,” *American Economic Review*, 106, 1144–1181.

AVERY, M., O. GIUNTELLA, AND P. JIAO (2025): “Why Don’t We Sleep Enough? A Field Experiment among College Students,” *Review of Economics and Statistics*, 107, 65–77.

BACHMANOV, A. A. AND G. K. BEAUCHAMP (2007): “Taste Receptor Genes,” *Annual Review of Nutrition*, 27, 389–414.

BAILEY, M. J., H. HOYNES, M. ROSSIN-SLATER, AND R. WALKER (2024): “Is the Social Safety Net a Long-Term Investment? Large-Scale Evidence from the Food Stamps Program,” *Review of Economic Studies*, 91, 1291–1330.

BARAHONA, N., C. OTERO, AND S. OTERO (2023): “Equilibrium Effects of Food Labeling Policies,” *Econometrica*, 91, 839–868.

BARKER, D. J. (1990): “The fetal and infant origins of adult disease,” *BMJ*, 301, 1111–1111.

BELLOC, M., F. DRAGO, M. FOCESATO, AND R. GALBIATI (2024): “Multigenerational Transmission of Wealth: Florence, 1403–1480,” *American Economic Journal: Applied Economics*, 16, 99–129.

BENCSEK, P., T. J. HALLIDAY, AND B. MAZUMDER (2023): “The intergenerational transmission of mental and physical health in the United Kingdom,” *Journal of Health Economics*, 92, 102805.

BIRCH, L. L. (1999): “Development of Food Preferences,” *Annual Review of Nutrition*, 19,

41–62.

- BLACK, S., N. DUZETT, A. LLERAS-MUNEY, N. POPE, AND J. PRICE (2024): “Intergenerational Transmission of Lifespan in the US,” *NBER Working Paper*, 31034.
- BLACK, S. E., P. J. DEVEREUX, AND K. G. SALVANES (2005): “Why the Apple Doesn’t Fall Far: Understanding Intergenerational Transmission of Human Capital,” *American Economic Review*, 95.
- BRATBERG, E., J. DAVIS, B. MAZUMDER, M. NYBOM, D. D. SCHNITZLEIN, AND K. VAAGE (2017): “A Comparison of Intergenerational Mobility Curves in Germany, Norway, Sweden, and the US,” *The Scandinavian Journal of Economics*, 119, 72–101.
- BRONNENBERG, B. J., J.-P. H. DUBÉ, AND M. GENTZKOW (2012): “The Evolution of Brand Preferences: Evidence from Consumer Migration,” *American Economic Review*, 102, 2472–2508.
- BÜCHEL, K., M. V. EHRLICH, D. PUGA, AND E. VILADECANS-MARSAL (2020): “Calling from the outside: The role of networks in residential mobility,” *Journal of Urban Economics*, 119, 103277.
- CALLAWAY, B. AND P. H. SANT’ANNA (2021): “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, 225, 200–230.
- CAMPBELL, J. AND J. COCHRANE (1999): “By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior,” *Journal of Political Economy*, 107, 205–251.
- CARROLL, C. D., J. OVERLAND, AND D. N. WEIL (2000): “Saving and Growth with Habit Formation,” *American Economic Review*, 90, 341–355.
- CHANG, H., T. J. HALLIDAY, M.-J. LIN, AND B. MAZUMDER (2024): “Estimating intergenerational health transmission in Taiwan with administrative health records,” *Journal of Public Economics*, 238, 2024.
- CHARLES, K. AND E. HURST (2003): “The Correlation of Wealth across Generations,” *Journal of Political Economy*, 111, 1155–1182.
- CHARNESS, G. AND U. GNEEZY (2009): “Incentives to Exercise,” *Econometrica*, 77, 909–931.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND B. MELLY (2013): “Inference on Counterfactual Distributions,” *Econometrica*, 81, 2205–2268.

- CHETTY, R., N. HENDREN, P. KLINE, E. SAEZ, AND N. TURNER (2014): “Is the United States Still a Land of Opportunity? Recent Trends in Intergenerational Mobility,” *American Economic Review*, 104, 141–147.
- CHUARD, P. AND V. GRASSI (2020): “Switzer-Land of Opportunity: Intergenerational Income Mobility in the Land of Vocational Education,” *Working Paper*.
- CLARK, G. AND N. CUMMINS (2015): “Intergenerational Wealth Mobility in England, 1858–1912: Surnames and Social Mobility,” *The Economic Journal*, 125, 61–85.
- COLE, J. B., J. C. FLOREZ, AND J. N. HIRSCHHORN (2020): “Comprehensive genomic analysis of dietary habits in UK Biobank identifies hundreds of genetic associations,” *Nature Communications*, 11, 1467.
- CORAK, M. (2020): “The Canadian Geography of Intergenerational Income Mobility,” *The Economic Journal*, 130, 2134–2174.
- DARDEN, M. AND D. GILLESKIE (2016): “The Effects of Parental Health Shocks on Adult Offspring Smoking Behavior and Self-Assessed Health,” *Health Economics*, 25, 939–954.
- DEUTSCHER, N. AND B. MAZUMDER (2020): “Intergenerational mobility across Australia and the stability of regional estimates,” *Labour Economics*, 66, 101861.
- (2023): “Measuring Intergenerational Income Mobility: A Synthesis of Approaches,” *Journal of Economic Literature*, 61, 988–1036.
- DICKSON, A., M. GEHRSTZ, AND J. KEMP (2023): “Does a Spoonful of Sugar Levy Help the Calories Go Down? An Analysis of the UK Soft Drinks Industry Levy,” *Review of Economics and Statistics*, 1–29.
- DUBOIS, P., R. GRIFFITH, AND M. O’CONNELL (2020): “How Well Targeted Are Soda Taxes?” *American Economic Review*, 110, 3661–3704.
- FADLON, I. AND T. H. NIELSEN (2019): “Family Health Behaviors,” *American Economic Review*, 109, 3162–3191.
- FORESTELL, C. A. (2024): “Does Maternal Diet Influence Future Infant Taste and Odor Preferences? A Critical Analysis,” *Annual Review of Nutrition*, 44.
- FORESTELL, C. A. AND J. A. MENNELLA (2007): “Early Determinants of Fruit and Vegetable Acceptance,” *Pediatrics*, 120, 1247–1254.

- FUHRER, J. C. (2000): “Habit Formation in Consumption and Its Implications for Monetary-Policy Models,” *American Economic Review*, 90, 367–390.
- GERVIS, J. E., J. MA, K. K. CHUI, N. M. McKEOWN, D. LEVY, AND A. H. LICHTENSTEIN (2023): “Bitter- and Umami-Related Genes are Differentially Associated with Food Group Intakes: the Framingham Heart Study,” *The Journal of Nutrition*, 153, 483–492.
- GOLDIN, J., T. HOMONOFF, AND K. MECKEL (2022): “Issuance and Incidence: SNAP Benefit Cycles and Grocery Prices,” *American Economic Journal: Economic Policy*, 14, 152–178.
- HALLIDAY, T. J., B. MAZUMDER, AND A. WONG (2020): “The intergenerational transmission of health in the United States: A latent variables analysis,” *Health Economics*, 29, 367–381.
- HASTINGS, J., R. KESSLER, AND J. M. SHAPIRO (2021): “The Effect of SNAP on the Composition of Purchased Foods: Evidence and Implications,” *American Economic Journal: Economic Policy*, 13, 277–315.
- HUT, S. (2020): “Determinants of Dietary Choice in the US: Evidence from Consumer Migration,” *Journal of Health Economics*, 72, 102327.
- HUT, S. AND E. OSTER (2022): “Changes in household diet: Determinants and predictability,” *Journal of Public Economics*, 208, 104620.
- JÄNTTI, M. AND S. P. JENKINS (2015): “Income Mobility,” *Handbook of Income Distribution*, 2, 807–935.
- KHERA, A. V., C. A. EMDIN, I. DRAKE, P. NATARAJAN, A. G. BICK, N. R. COOK, D. I. CHASMAN, U. BABER, R. MEHRAN, D. J. RADER, V. FUSTER, E. BOERWINKLE, O. MELANDER, M. ORHO-MELANDER, P. M. RIDKER, AND S. KATHIRESAN (2016): “Genetic Risk, Adherence to a Healthy Lifestyle, and Coronary Disease,” *New England Journal of Medicine*, 375, 2349–2358.
- KIESEL, K., H. LANG, AND R. J. SEXTON (2023): “A New Wave of Sugar-Sweetened Beverage Taxes: Are They Meeting Policy Goals and Can We Do Better?” *Annual Review of Resource Economics*, 15, 407–432.
- LI, C. AND D. B. GILLESKIE (2021): “The influence of endogenous behaviors among social pairs: Social interaction effects of smoking,” *Journal of Health Economics*, 80, 102547.
- MAZUMDER, B. (2024): “What do we know about the intergenerational transmission of health,”

Research Handbook on Intergenerational Inequality, 150-163.

- MENNELLA, J. A., N. K. BOBOWSKI, AND D. R. REED (2016): "The development of sweet taste: From biology to hedonics," *Reviews in Endocrine and Metabolic Disorders*, 17, 171–178.
- MENNELLA, J. A., C. P. JAGNOW, AND G. K. BEAUCHAMP (2001): "Prenatal and Postnatal Flavor Learning by Human Infants," *Pediatrics*, 107, e88–e88.
- MENNELLA, J. A., M. Y. PEPINO, AND D. R. REED (2005): "Genetic and Environmental Determinants of Bitter Perception and Sweet Preferences," *Pediatrics*, 115, e216–e222.
- OSTER, E. (2018): "Diabetes and Diet: Purchasing Behavior Change in Response to Health Information," *American Economic Journal: Applied Economics*, 10, 308–348.
- SCAGLIONI, S., V. DE COSMI, V. CIAPPOLINO, F. PARAZZINI, P. BRAMBILLA, AND C. AGOSTONI (2018): "Factors Influencing Children's Eating Behaviours," *Nutrients*, 10.
- SCHILBACH, F. (2019): "Alcohol and Self-Control: A Field Experiment in India," *American Economic Review*, 109, 1290–1322.
- STORY, M., K. M. KAPHINGST, R. ROBINSON-O'BRIEN, AND K. GLANZ (2008): "Creating Healthy Food and Eating Environments: Policy and Environmental Approaches," *Annual Review of Public Health*, 29, 253–272.
- SØBERG, S., C. H. SANDHOLT, N. Z. JESPERSEN, U. TOFT, A. L. MADSEN, S. VON HOLSTEIN-RATHLOU, T. J. GREVENGOED, K. B. CHRISTENSEN, W. L. BREDIE, M. J. POTTHOFF, T. P. SOLOMON, C. SCHEELE, A. LINNEBERG, T. JØRGENSEN, O. PEDERSEN, T. HANSEN, M. P. GILLUM, AND N. GRARUP (2017): "FGF21 Is a Sugar-Induced Hormone Associated with Sweet Intake and Preference in Humans," *Cell Metabolism*, 25, 1045–1053.e6.
- TOSI, F. AND R. RETTAROLI (2022): "Intergenerational transmission of dietary habits among Italian children and adolescents," *Economics & Human Biology*, 44, 101073.
- VERGHESE, A., M. RABER, AND S. SHARMA (2019): "Interventions targeting diet quality of Supplemental Nutrition Assistance Program (SNAP) participants: A scoping review," *Preventive Medicine*, 119, 77–86.
- WISE, P. M., L. NATTRESS, L. J. FLAMMER, AND G. K. BEAUCHAMP (2016): "Reduced dietary intake of simple sugars alters perceived sweet taste intensity but not perceived pleasantness," *The American Journal of Clinical Nutrition*, 103, 50–60.

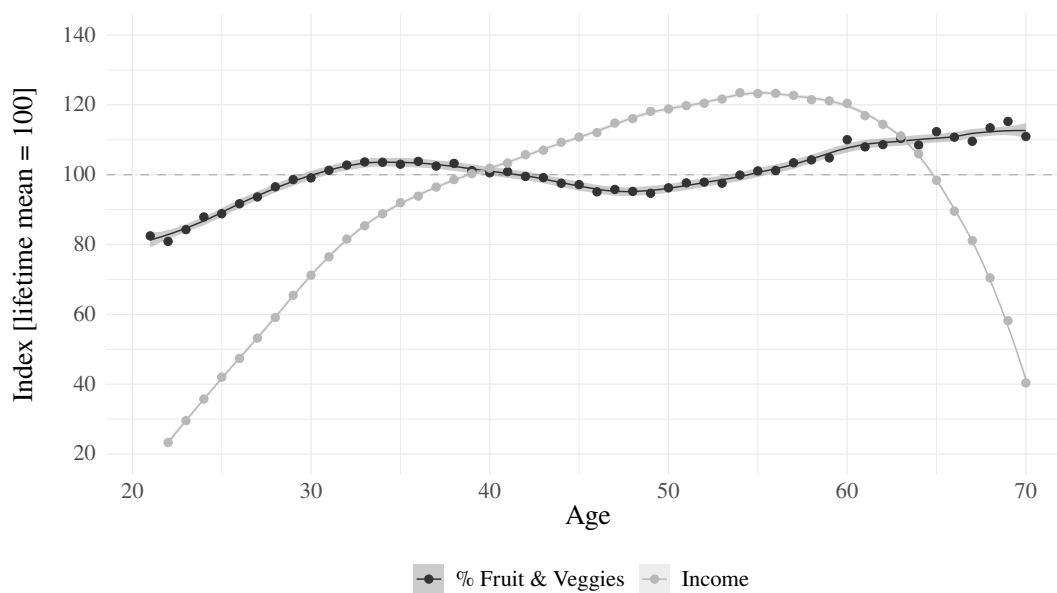
A Data: Additional Summary Statistics

Table A1: Summary Statistics for Children

Panel a)	Final Sample (N = 270,957)			Population (N = 2,276,806)	
	Mean	Wt. Mean	SD	Mean	SD
Age	43.72	43.85	10.69	43.70	11.70
Age Father	71.87	71.79	9.66	71.05	10.35
Age Mother	71.03	71.21	10.35	70.85	11.36
HH Income	142.37	131.53	137.04	129.68	109.09
HH Income Adjusted	83.25	83.01	87.01	81.60	64.88
Panel b)	Share	Wt. Share	N	Share	N
<i>Gender</i>			270,957		2,276,806
Female	0.539	0.545	146,148	0.508	1,155,646
Male	0.461	0.455	124,809	0.492	1,121,160
<i>Marriage</i>			270,957		2,276,806
Married	0.623	0.517	168,776	0.503	1,145,736
Not Married	0.377	0.483	102,181	0.497	1,131,070
<i>Highest Education</i>			193,711		1,554,739
Tertiary	0.535	0.502	103,676	0.500	777,901
Secondary	0.427	0.451	82,763	0.446	694,110
Primary	0.038	0.048	7,272	0.053	82,728
<i>Language Region</i>			270,711		2,274,341
German	0.769	0.806	208,283	0.723	1,644,202
French	0.191	0.170	51,643	0.220	500,133
Italian	0.040	0.025	10,785	0.057	130,006
<i>Pop. Density</i>			270,711		2,274,341
Rural	0.217	0.216	58,732	0.216	490,681
Suburban	0.582	0.522	157,660	0.522	1,186,301
Urban	0.201	0.263	54,319	0.263	597,359
<i>Household Size</i>			270,957		2,276,806
1	0.102	0.210	27,715	0.210	478,435
2	0.269	0.332	72,900	0.332	754,928
3-4	0.511	0.372	138,377	0.372	846,201
5+	0.118	0.087	31,965	0.087	197,242

Notes: This table presents summary statistics for the children in the final dataset alongside summary statistics for the relevant population. Columns 1 and 3 report the mean and standard deviation (SD) for the final sample, while Columns 4 and 5 present the corresponding values for the population. Column 2 shows the results after reweighting the sample to match the population distribution of household size and population density. *HH Income Total* represents a household's average annual gross labor market income for the period 2016–2020, measured in 1,000 CHF. *HH Income Adjusted* is income adjusted by the square root of household size. All other variables reflect their values for the year 2020. *Highest Education* indicates the highest level of education completed by any household member, and *Pop. Density* refers to the municipality's population density.

Figure 1: Life Cycle in Income and Diet



Notes: The figure shows the average of two variables for each age group between 21 and 70: (i) the annual household income in the target population adjusted by the square root of household members (3.1 million observations) and (ii) the expenditure share on fruit and vegetables in the sample (270,957 observations). All values are normalized to 100 based on the lifetime average of each variable. The conditional expectation functions, together with uniform 95% confidence bands, are estimated using a local regression, with weights proportional to the size of the age groups.

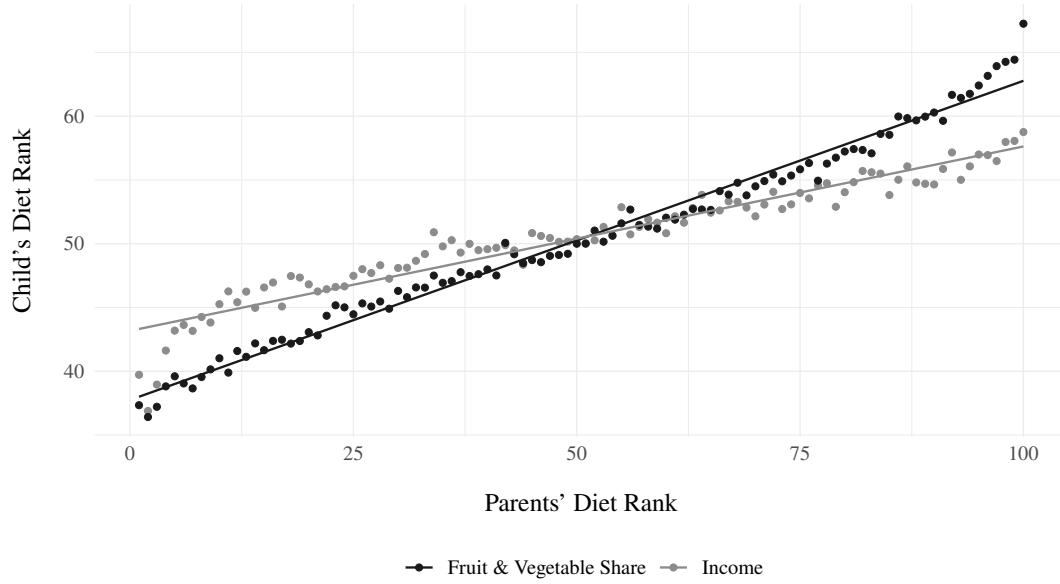
Figure 2: Transition Matrix

Child's Produce Consumption Quintile	5	11.5 % [11.2, 11.7]	15.0 % [14.7, 15.3]	18.4 % [18.1, 18.7]	23.2 % [22.9, 23.5]	31.9 % [31.6, 32.2]
	4	15.0 % [14.7, 15.3]	18.3 % [18.0, 18.6]	20.6 % [20.3, 20.9]	22.4 % [22.1, 22.7]	23.8 % [23.5, 24.1]
	3	18.8 % [18.5, 19.1]	20.5 % [20.2, 20.8]	21.4 % [21.1, 21.8]	20.8 % [20.5, 21.1]	18.4 % [18.1, 18.7]
	2	23.5 % [23.1, 23.8]	22.9 % [22.6, 23.2]	20.7 % [20.4, 21.0]	18.4 % [18.1, 18.7]	14.5 % [14.3, 14.8]
	1	31.3 % [30.9, 31.6]	23.3 % [23.0, 23.6]	19.0 % [18.6, 19.3]	15.2 % [14.9, 15.5]	11.4 % [11.1, 11.6]
		1	2	3	4	5
		Parents' Produce Consumption Quintile				

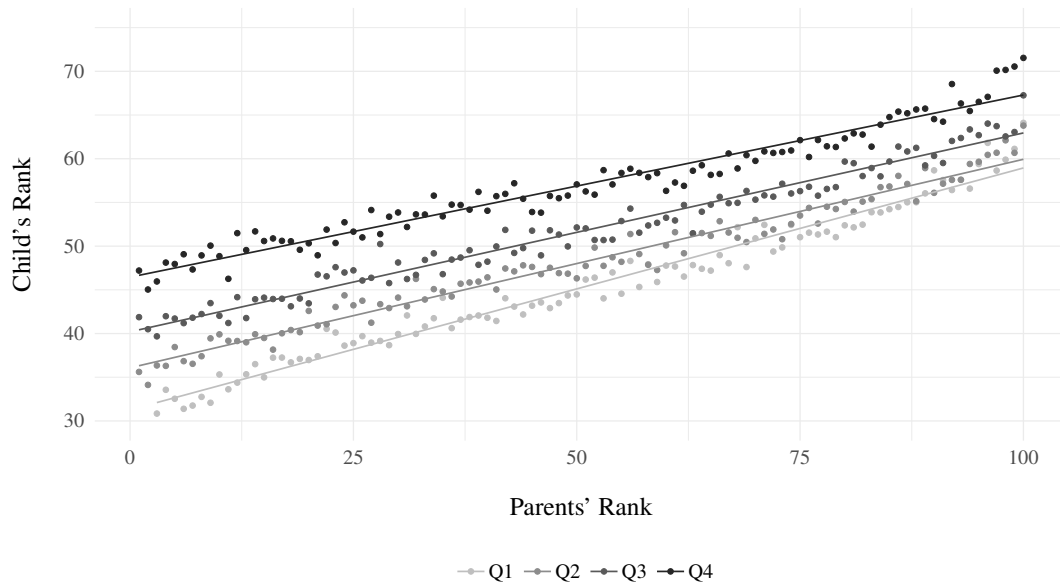
Notes: The figure shows the transition probabilities for children's diet conditional on their parents' diet (Equation 4). We analyze transitions between quintiles of the distribution and calculate the quintiles on age groups within the respective sub-sample of children and parents. 95% confidence intervals in parentheses are estimated using 1,000 bootstrap replications.

Figure 3: Rank-Rank Slopes

(a) RRS of Diet and Income



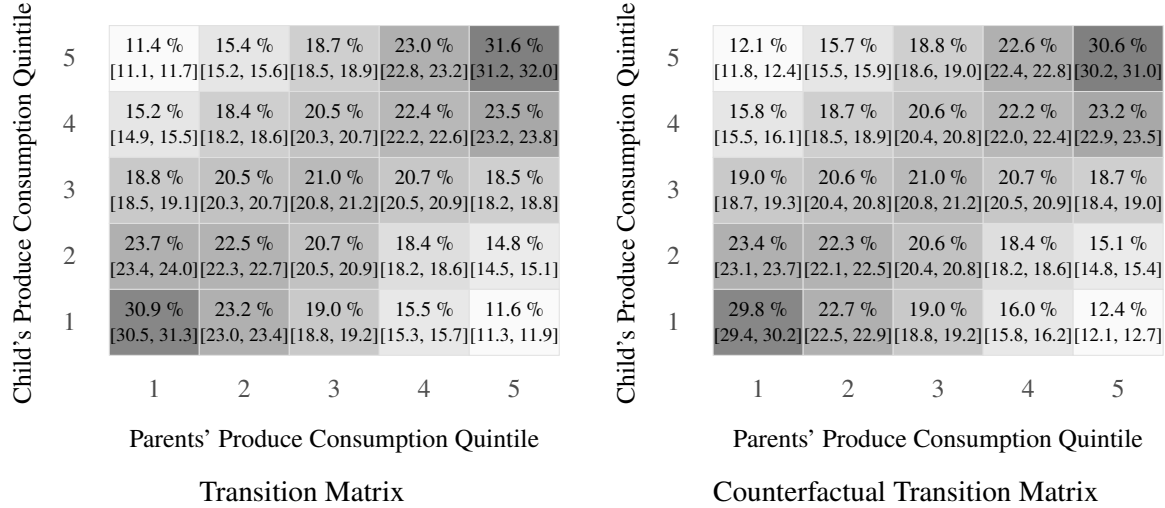
(b) RRS of Diet by Income Quartiles



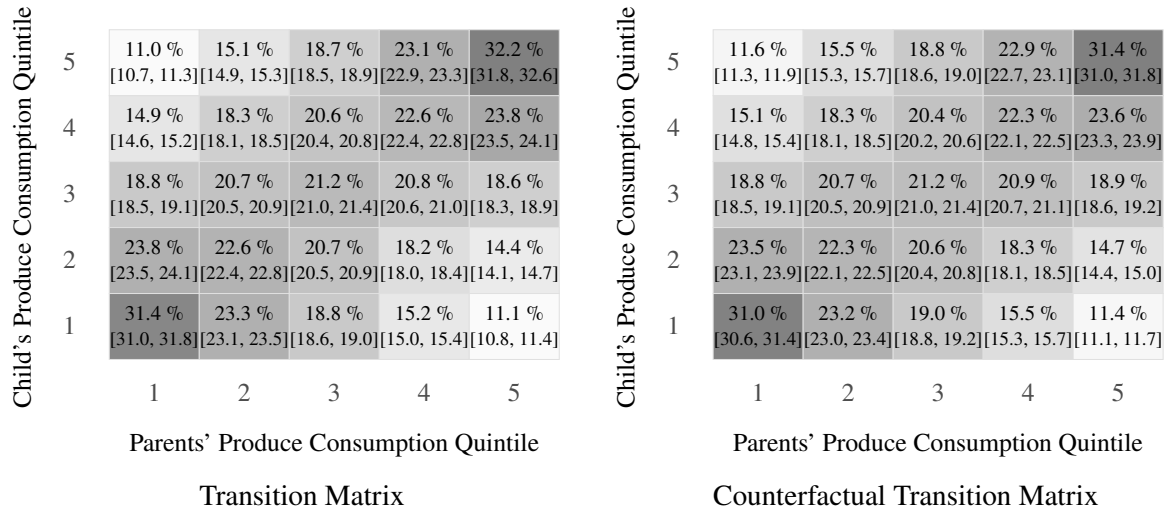
Notes: Panel (a) shows the estimation results for the rank-rank regression in Equation (1) for diet and income. Panel (b) shows estimation results for the rank-rank regression of diet for the four income quartiles. The dots represent estimates of the conditional expectation of children's percentile ranks at each parental percentile rank.

Figure 4: Transition Matrices: the Role of Income, Education, and Location

(a) Role of Income and Education

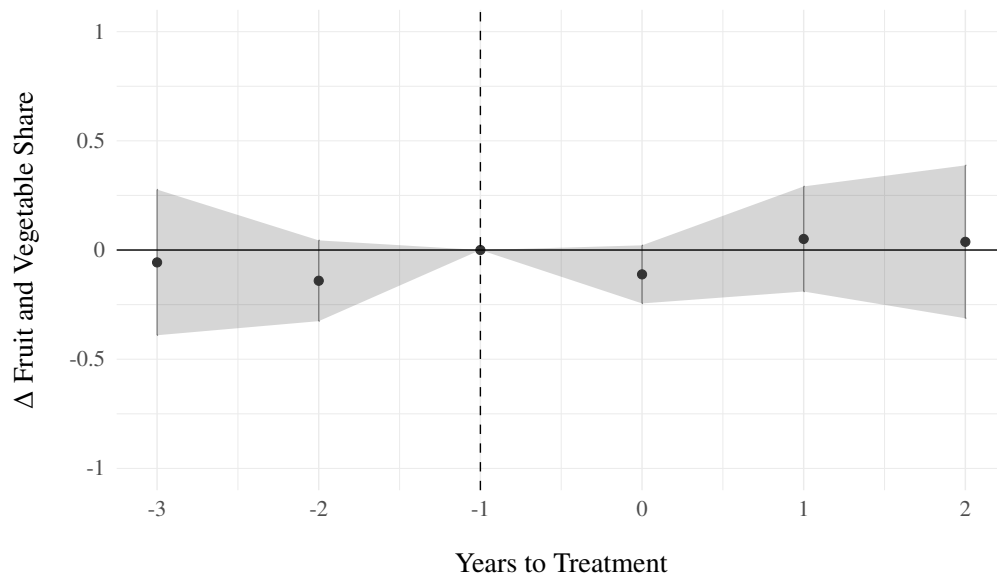


(b) Role of Location



Notes: Panel (a) shows the transition matrix (left) and the counterfactual transition matrix (right) for income and education. In the counterfactual scenario, children's income and education are assigned independently from their parents' values. The results are estimated using the sample of 135,213 children for whom we observe both their own and their parents' education. Panel (b) shows the transition matrix (left) and the counterfactual transition matrix (right) for location. In the counterfactual scenario, children's locations are assigned independently of their parents' values. The results are estimated using the sample of 120,424 children for whom we observe both their own and their parents' residence location and education. 95% confidence intervals in parentheses are estimated using 1,000 bootstrap replications.

Figure 5: Effect of the Lifestyle-Related Death of a Parent



Notes: The figure shows difference-in-differences estimates of the effect of a lifestyle-related parental death on children's annual fruit and vegetable intake using the estimator suggested by [Callaway and Sant'Anna \(2021\)](#). We use the *not-yet-treated* units as the comparison group. The estimation uses 35,073 observations, coefficients are normalized to the year before the treatment, and standard errors are clustered at the individual level.

Table 1: Summary Statistics for Children's Expenditures

	Monthly Spending			% Fruit & Vegetable			Budget Survey	
	Mean	p50	SD	Mean	p50	SD	Spending	Share
<i>Overall</i>	399	323	284	0.15	0.14	0.07	616	0.65
<i>By Age</i>								
< 34	298	239	207	0.15	0.14	0.07	459	0.65
35–44	425	357	287	0.15	0.14	0.07	654	0.65
45–54	459	382	316	0.14	0.14	0.07	728	0.63
55–64	393	325	274	0.16	0.15	0.08	663	0.59
65+	345	286	237	0.17	0.16	0.08	616	0.56
<i>By Household Income</i>								
1st Quintile	269	212	191	0.14	0.13	0.08	409	0.66
2nd Quintile	294	230	215	0.14	0.13	0.08	485	0.61
3rd Quintile	374	305	259	0.14	0.13	0.07	604	0.62
4th Quintile	422	354	283	0.15	0.14	0.06	713	0.59
5th Quintile	458	384	312	0.16	0.16	0.07	869	0.53
<i>By Highest Education</i>								
Primary	275	222	190	0.13	0.12	0.07		
Secondary	376	304	264	0.14	0.13	0.07		
Tertiary	442	368	303	0.16	0.16	0.07		
<i>By Pop. Density</i>								
Rural	386	317	266	0.14	0.13	0.06		
Suburban	407	332	288	0.15	0.14	0.07		
Urban	389	303	289	0.17	0.16	0.08		

Notes: The table presents summary statistics for the transaction records of food expenditures in our final dataset (N = 270,957). In the columns titled *Budget Survey*, we compare these figures with the average monthly grocery expenditures for food and beverages reported in the administrative Household Budget Surveys for the years 2015 to 2017 (N = 9,955). The last column shows the average monthly expenditures in our data relative to the survey. Quintiles of *Household Income* are computed using a household's average gross labor market income for the period 2016–2020. The budget survey provides summary statistics for each income quintile. All other variables reflect their values for the year 2020. *Highest Education* represents the highest level of education completed by any household member, and *Pop. Density* refers to the population density of the municipality where the individual resides.

Table 2: Main Mobility Measures

	(a) Rank-Rank Reg.		(b) IGE	(c) CER		(d) Transition Prob.		
	Intercept	Slope	AME	25	75	Q1Q1	Q1Q5	Q5Q5
Diet	37.75 (0.09)	0.250 (0.002)	0.247 (0.002)	44.97 (0.68)	56.07 (0.66)	31.26 (0.17)	11.47 (0.13)	31.89 (0.17)
Income	43.17 (0.13)	0.145 (0.003)	0.113 (0.002)	47.48 (0.78)	53.98 (0.83)	26.43 (0.22)	14.14 (0.19)	28.60 (0.21)

Notes: The diet results are estimated using 270,957 observations. The income results are estimated using 161,504 observations and we restrict the sample to children between 32 and 38 and fathers between 50 and 62. The IGE for diet gives the average marginal effect for a quadratic specification and the IGE for income uses the log of the father's income as an explanatory variable and the log of the children's income as a dependent variable. Standard errors are computed using 1,000 bootstrap replications.

Table 3: Heterogeneities

	Rank-Rank		IGE	CER		Transition Prob.			N
	RRS	P-value	AME	25	75	Q1Q5	Q1Q1	Q5Q5	
<i>(a) Child's Education</i>									
Primary	0.242 (0.012)	0.446	0.247 (0.014)	39.044 (3.745)	49.192 (4.364)	7.66 (0.68)	43.78 (1.22)	24.16 (1.18)	7,272
Secondary	0.234 (0.003)		0.222 (0.004)	40.347 (1.118)	50.766 (1.329)	8.18 (0.21)	37.68 (0.34)	24.61 (0.34)	82,763
Tertiary	0.229 (0.003)		0.223 (0.003)	49.002 (1.089)	60.652 (1.005)	15.46 (0.27)	23.43 (0.30)	36.66 (0.29)	103,676
<i>(b) Child's Income</i>									
1st Quartile	0.279 (0.004)	0.000	0.268 (0.004)	39.874 (1.271)	51.591 (1.462)	8.00 (0.22)	40.98 (0.40)	28.46 (0.40)	64,881
2nd Quartile	0.239 (0.004)		0.226 (0.004)	42.772 (1.279)	52.301 (1.423)	9.22 (0.26)	32.83 (0.40)	27.07 (0.39)	64,873
3rd Quartile	0.228 (0.004)		0.220 (0.004)	46.784 (1.362)	55.771 (1.318)	12.04 (0.28)	27.91 (0.40)	31.40 (0.41)	64,863
4th Quartile	0.208 (0.004)		0.214 (0.005)	51.812 (1.495)	61.904 (1.203)	18.18 (0.37)	20.86 (0.41)	38.73 (0.38)	64,852
<i>(c) Child's place of residence</i>									
Rural	0.236 (0.004)	0.009	0.228 (0.004)	42.015 (1.367)	51.837 (1.456)	8.02 (0.24)	35.43 (0.39)	24.67 (0.44)	58,732
Suburban	0.235 (0.002)		0.230 (0.003)	43.952 (0.870)	55.093 (0.870)	10.98 (0.17)	31.57 (0.24)	29.34 (0.24)	157,660
Urban	0.221 (0.004)		0.224 (0.006)	53.833 (1.581)	62.738 (1.358)	20.28 (0.46)	22.80 (0.50)	41.82 (0.40)	54,319
<i>(d) Distance to Parents</i>									
1st Quartile	0.281 (0.004)	0.000	0.273 (0.004)	41.547 (1.294)	55.697 (1.455)	8.57 (0.25)	34.77 (0.40)	30.60 (0.41)	63,842
2nd Quartile	0.252 (0.004)		0.244 (0.004)	44.069 (1.375)	54.820 (1.311)	10.44 (0.27)	31.43 (0.40)	30.71 (0.42)	63,841
3rd Quartile	0.225 (0.004)		0.222 (0.004)	43.804 (1.356)	53.989 (1.341)	12.88 (0.29)	29.90 (0.40)	30.77 (0.41)	63,841
4th Quartile	0.202 (0.004)		0.204 (0.005)	49.483 (1.444)	57.584 (1.309)	15.34 (0.33)	26.68 (0.39)	33.52 (0.37)	63,841
<i>(e) Nationality</i>									
Swiss	0.251 (0.002)	0.052	0.248 (0.002)	45.141 (0.709)	56.124 (0.674)	11.46 (0.13)	31.31 (0.18)	31.89 (0.17)	267,421
Foreign	0.217 (0.017)		0.209 (0.023)	39.556 (6.171)	59.714 (5.160)	10.84 (1.39)	29.45 (2.08)	29.55 (1.41)	3,536

Notes: The table shows the results for different subsamples defined by education, income, residence, distance to their parents, and national origin. The second column gives the p -value of the null hypothesis that the rank-rank slope is the same for all subgroups. Bootstrap standard errors in parentheses are computed using 1,000 replications. The number of observations in each subgroup is shown in the last column.

Table 4: ATT of Lifestyle-Related Death of a Parent on Dietary Behavior

	Fruit & Veggies	Meat & Fish	Convenience	Eggs & Dairy
Treatment effect	-0.008 (0.111)	0.177 (0.148)	-0.285 (0.106)	0.161 (0.110)
Pre-treatment mean	15.8	23.5	12.4	18.1

Notes: The table shows difference-in-differences estimates of the lifestyle-related death of a parent's effect on their children's annual expenditure on different product categories using the estimator suggested by [Callaway and Sant'Anna \(2021\)](#). We use the *not-yet-treated* units as the comparison group. Standard errors are clustered at the individual level.

Table 5: Average Marginal Effect for Children with Divorced Parents

Child Age at Divorce:	Fruit & Vegetable Share Child					
	≤ 10 (1)	10–18 (2)	18–25 (3)	≤ 10 (4)	10–18 (5)	18–25 (6)
Fruit & Vegetable Share Mother	0.220 (0.018)	0.255 (0.014)	0.229 (0.014)			
Fruit & Vegetable Share Father				0.159 (0.024)	0.138 (0.025)	0.154 (0.025)
Observations	3,149	5,203	4,913	1,273	1,254	1,523

Notes: The table shows the average marginal effect of mothers' and fathers' diets separately for the subsample of children with divorced parents who never remarried. We estimate separate regressions of the child's fruit and vegetable share on each parent's share and its square. Further, we control for the parents' and child's age as well as their squares. Standard errors are computed using 1,000 bootstrap replications.